Ken Nakamura

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RESEARCH FOCUS

I am interested in how human cognition and conceptual understanding arise: how perceptual experience is organized into concepts, how symbolic systems such as language emerge and become grounded, and what mathematical principles govern these processes. I approach these questions through NeuroAI, model–brain alignment, representation analysis, and self-supervised learning.

EDUCATION

The University of Tokyo, Faculty of Engineering — Apr 2023–Mar 2027 (expected)
B.Eng., Department of Mathematical Engineering and Information Physics — *GPA: 4.11/4.30*

- Admitted through UTokyo's School Recommendation-Based Selection; one of about 30 Faculty of Engineering admits per year (approximately 1.0% of UTokyo's undergraduate intake).
- Permitted to exceed UTokyo's 30-credit-per-semester registration cap based on first-semester academic performance (approximately 10 students per cohort).

PUBLICATIONS AND PREPRINTS

[1] **Nakamura, K.**, Nakai, T., Yashiro, R., Yamashita, A., & Amano, K. (2026). Beyond Prediction Accuracy: Target-Space Recovery Profiles for Evaluating Model-Brain Alignment. *arXiv preprint arXiv:2605.20127*. Under review.

[2] Asanuma, H., Koide-Majima, N., **Nakamura, K.**, Horii, T., Nishimoto, S., & Oizumi, M. (2025). Correspondence of high dimensional emotion structures elicited from video clips between humans and multimodal LLMs. *Scientific Reports*, 15(1), 32175.

RESEARCH EXPERIENCE

Undergraduate Researcher — Jun 2024–Present
Amano-Nakai-Nakayama Lab, Department of Information Physics and Computing, Graduate School of Information Science and Technology, The University of Tokyo — *Bunkyo-ku, Tokyo, Japan*
Advisor: Prof. Kaoru Amano, Assoc. Prof. Tomoya Nakai

- Study how learned representations capture perceptual and conceptual structure in human brain responses, beyond aggregate prediction accuracy.
- Proposed target-space recovery profiles for identifying which reproducible dimensions of brain-response space are captured by models and other human brains; led a first-author preprint.

Undergraduate Researcher — Apr 2024–Aug 2025
Oizumi Lab, Department of General Systems Studies, Graduate School of Arts and Sciences, The University of Tokyo — *Meguro-ku, Tokyo, Japan*
Advisor: Prof. Masafumi Oizumi

- Compared the representational geometry of multimodal LLM-inferred and human-reported emotion spaces elicited by naturalistic videos.
- Analyzed item- and category-level model–human alignment using RSA and Gromov-Wasserstein optimal transport; co-authored a peer-reviewed publication.

Guest Researcher — Feb 2026–Apr 2026
Smart Data & Knowledge Services Department, German Research Center for Artificial Intelligence (DFKI) — *Kaiserslautern, Germany*
Advisor: Prof. Dr. Prof. h.c. Andreas Dengel

- Studied continual LLM self-distillation under controlled tool-use and mathematical-reasoning settings, focusing on when self-training destabilizes representation and task-learning dynamics.
- Built unified training and evaluation protocols and presented the findings in a DFKI research webinar.

Visiting Research Fellow Aug 2025–Sep 2025
Sulchek BioMEMS and Biomechanics Lab, Georgia Institute of Technology *Atlanta, Georgia, USA*
Advisor: Prof. Todd Sulchek

- Curated over 23,000 labeled cell-morphology images from high-speed microfluidic compression videos and trained lightweight computer-vision models for abnormal morphology detection.

Undergraduate Research Assistant Oct 2023–Jan 2024
Ikeuchi Lab, Department of Biomolecular and Cellular Engineering, Institute of Industrial Science, The University of Tokyo *Meguro-ku, Tokyo, Japan*
Advisor: Prof. Yoshiho Ikeuchi

- Assisted experiments using electrical feedback to train in vitro mouse cerebral networks on maze-navigation tasks; contributed to stimulation design and data collection.

RESEARCH PRESENTATIONS

[1] **Nakamura, K.**, & Frolov, S. (Apr 2026). *Toward Efficient Continual Learning for LLMs: Two Burdens of Self-Distillation*. Webinar presented at the Smart Data & Knowledge Services Department, German Research Center for Artificial Intelligence (DFKI), Kaiserslautern, Germany.

[2] **Nakamura, K.**, Amiri, H. A., Chen, S., & Sulchek, T. (Sep 2025). *BlebNet: a Cell Damage Detection Network for High-speed Microfluidic Compressions*. Poster presented at the Nakatani RIES Symposium, Georgia Institute of Technology, Atlanta, Georgia, USA.

[3] **Nakamura, K.**, Asanuma, H., & Oizumi, M. (Sep 2024). *Emotion Prediction During Video Viewing Using Multimodal LLMs*. Final seminar presentation, The University of Tokyo, Meguro-ku, Tokyo, Japan.

FELLOWSHIPS AND GRANTS

MITOU IT Program, Selected Developer 2025–2026
Ministry of Economy, Trade and Industry (METI), Government of Japan

- Selected as one of 21 projects from 227 applications (approximately 9.3%); received JPY 2.88 million (approximately USD 18,000) in project funding.

Nakatani Research and International Experiences for Students Fellowship Program Mar 2025
Nakatani Foundation / Georgia Institute of Technology

- Selected as one of 11 Japanese fellows from 130 applicants (approximately 8.5%); supported a fully funded research stay at Georgia Institute of Technology.

Nakatani Advanced Research Internship Program Oct 2025
Nakatani Foundation

- Selected as one of 5 Japanese students for a fully funded research stay at DFKI.

ASJ Foundation Scholarship Jun 2025
ASJ Foundation

- Selected as one of 12 undergraduate recipients in 2025.

KEYENCE Foundation Student Support Grant Jun 2026
KEYENCE Foundation

HONORS AND AWARDS

MITOU Super Creator Certification (Press) Jun 2026
Ministry of Economy, Trade and Industry (METI), Government of Japan

- Certified as one of 23 MITOU Super Creators among 34 creators who completed the 2025 MITOU IT Program.

Chemistry Grand Prix Awards 2021–2022
Chemical Society of Japan

- Bronze Prize (Aug 2022): awarded in Japan's national chemistry competition for high-school-and-younger students; approximately top 2.5% among 3,215 participants.
- Kanto Branch Chief's Award (Nov 2021), Chemical Society of Japan, Kanto Branch.

Valedictorian Mar 2023
Seiko Gakuin Junior and Senior High School

TECHNICAL SKILLS

Programming	Python, C/C++, MATLAB, Rust, JavaScript/TypeScript, LaTeX
Machine Learning & Representation Analysis	PyTorch, TensorFlow, computer vision, fMRI encoding analysis, model–brain alignment, optimal transport, multimodal LLM evaluation, continual learning
Research & Software Infrastructure	Docker, Git, SSH, dataset curation and annotation, React, Next.js, Vite, command-line tools

LANGUAGES

Japanese (Native), English (Fluent, TOEFL iBT 99/120)